FACULTY OF COMPUTING

SEMESTER II

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SECP 3843 – SPECIAL TOPIC IN DATA ENGINEERING

SECTION 02

Tutorial: Article

LECTURER: DR. ARYATI BINTI BAKRI

NAME	MATRIC NO
NUR FIRZANA BINTI BADRUS HISHAM	A23CS0156
NURAISYAH BINTI MOHD ZIKRE	A23CS0160

What is the Main Problem?

The problem identified from this article is PT. MPM, a regional product packaging manufacturer, lacks an appropriate system to properly measure and monitor its core business processes (sales and production) which cause the top management to make decisions ineffectively and reduce the company's operational efficiency and its market competitiveness against rivals' competitiveness.

What is the Main Solution?

The solution proposed and implemented in this research is the development of Business Intelligence (BI) Dashboard System for PT. MPM using the Waterfall methodology and Google Data Studio as a visualization tool. The solution directly identified the problems with three components.

- Structure performance visibility with BI dashboards

  To solve the lack of monitoring system, two custom dashboards were created; Sales Performance Dashboard and Production Performance Dashboard. These dashboards provide management with a single picture of the company’s most important processes, as opposed to the absence of structured performance information that was previously present.
- A controlled data pipeline for reliable data

  If the data is not reliable then any dashboard created would be ineffective, so the solution is by using the staging ETL (Extract, Transform, Load) pipeline. This pipeline ensures all the data on the dashboards is accurate and consistent and directly minimizes the ineffective decision making.
- Top Management interactive self-service decision support

  The dashboard was developed to adopt the smart and interactive dashboard principles and made self-service BI tools. The system can be accessed by anyone with a web browser, allowing decision makers to access these analytical capabilities without having to wait for reports to be compiled manually and sent to them.

User Story

Based on the system requirements and the Data Flow Diagram (DFD) user roles identified in the study:

- As Top Management, I want to view a centralized dashboard that is able to display sales KPIs (revenue, forecast vs actual, product categories) so I can evaluate sales performance and make strategic decisions..
- As a Data Analyst, I want to upload, update data and perform extract, transform, and load (ETL) raw reports into the data warehouse so that clean, accurate data feeds into the BI dashboards that always reflect current data.
- As a Production Division Manager, I want to monitor OEE (Overall Equipment Effectiveness), machine availability, and production output vs plan.
- As a System Admin, I want to edit the layout of the dashboard and manage the user access to decide who can view the dashboard.

Entity Relationship Diagram (ERD)

From the article, Entity Relationship Diagram (ERD) is defined as in *Figure 1* where the diagram structures the transactional data layer that the paper describes flowing into the core system before warehouse distribution.

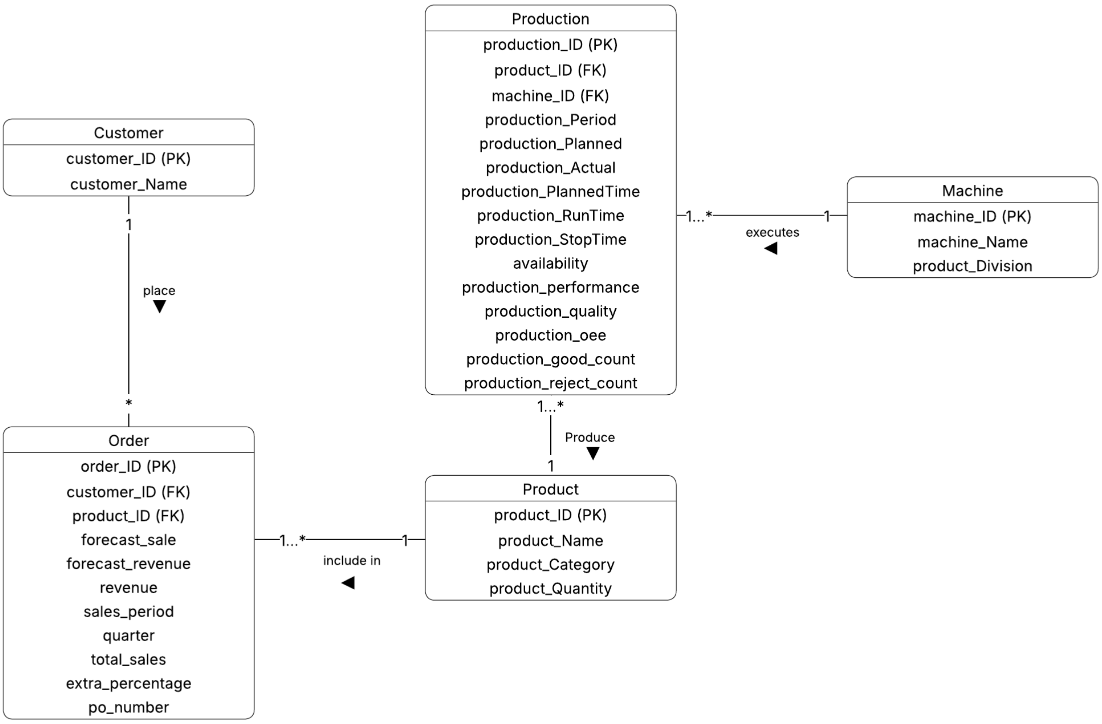

Figure 1: Physical Entity Relationship Diagram before warehouse distribution

Use Case Diagram

Use Case diagram is used to answer what in the PT. MPM dashboard system. It documented all people or actors that involved in the Business Intelligence (BI) dashboard system and all the events (use cases) that can perform and connect the both together as shown in Figure 2.

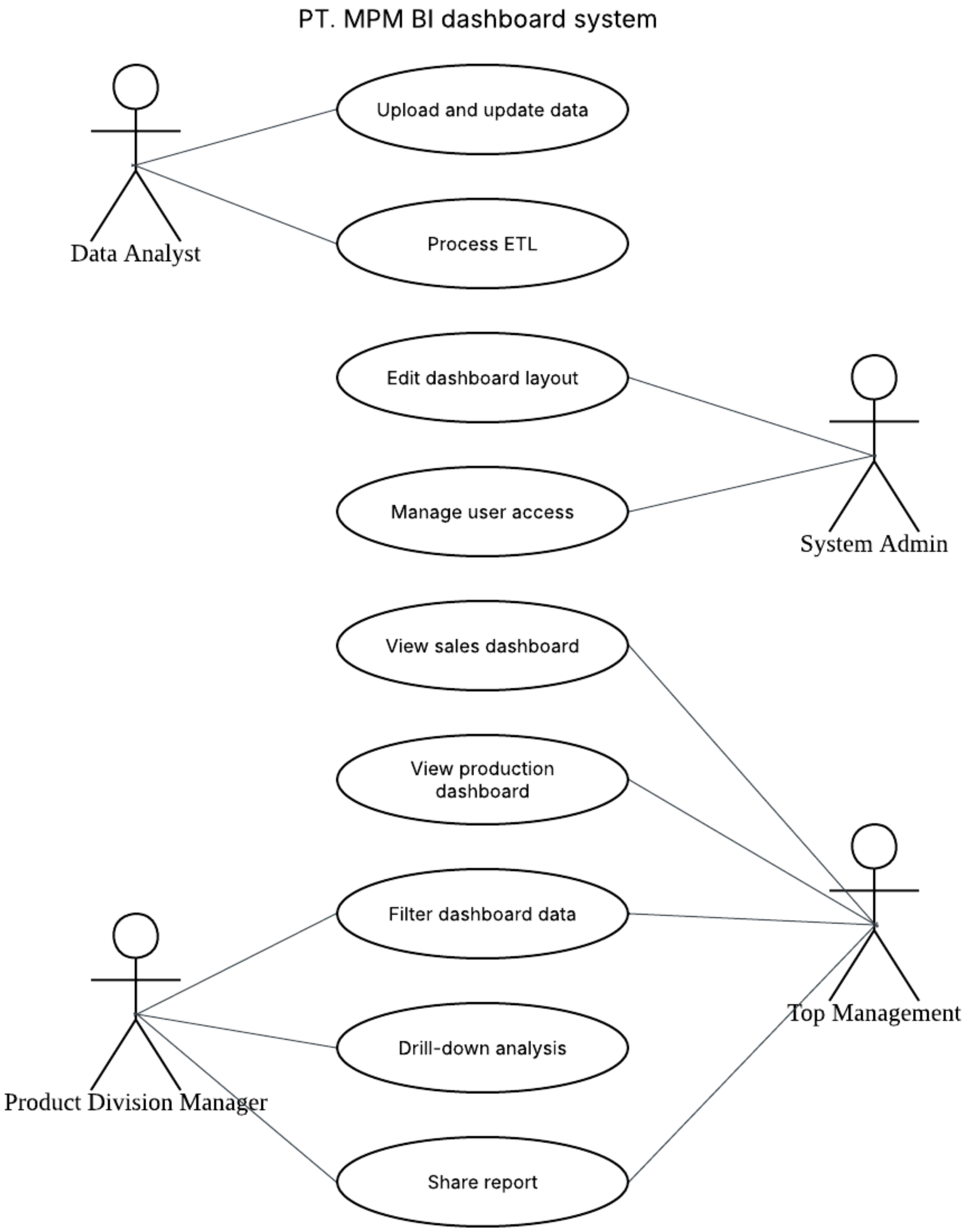

Figure 2: Use Case Diagram for PT. MPM BI Dashboard System

Data Architecture

The 4-layer data architecture shown in Figure 3 illustrates how the raw data sources is collected, extracted and processed through various cleaning stages to present useful information to top management.

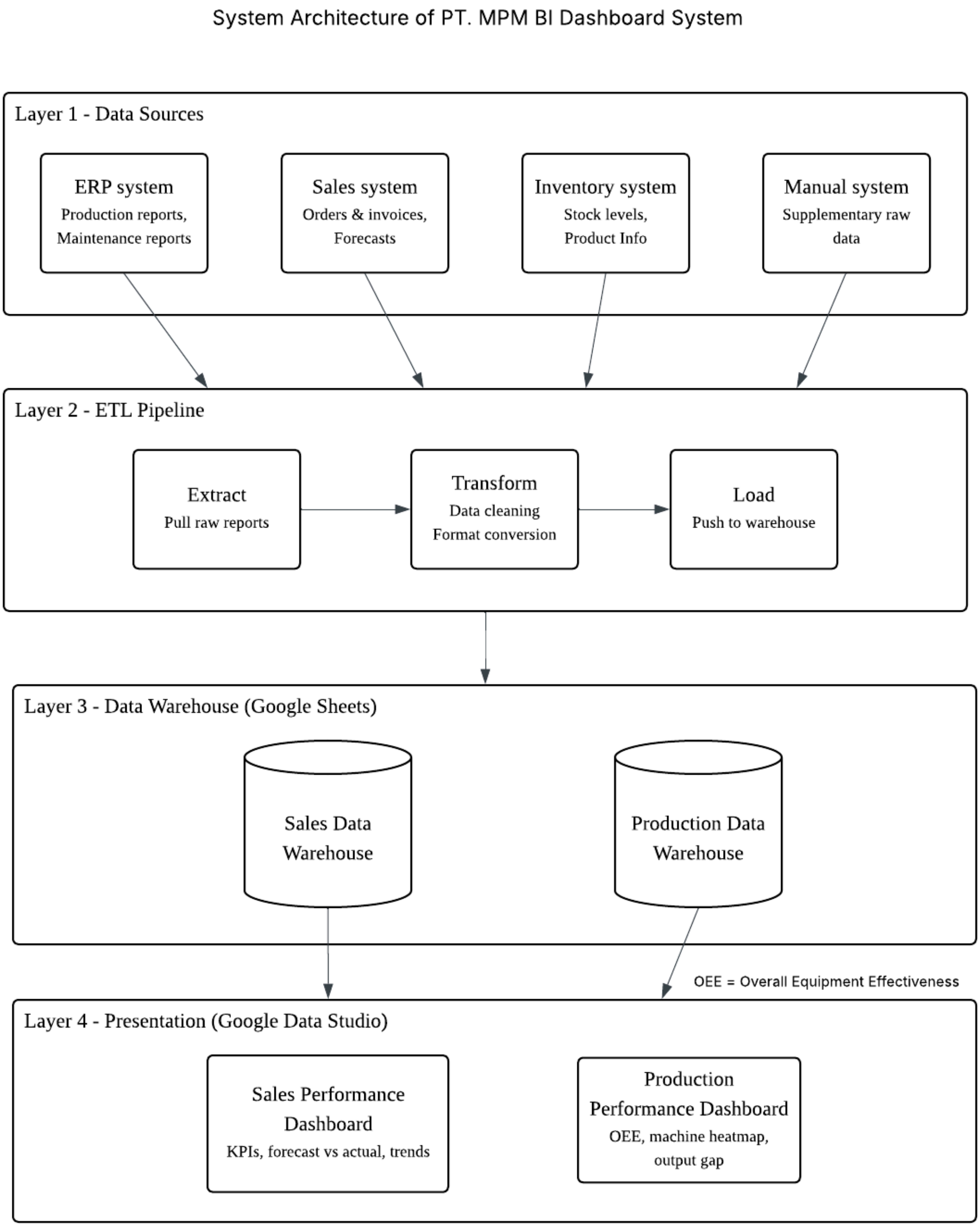

Figure 3: System Architecture for PT. MPM BI Dashboard System

Multidimensional Galaxy Schema

The galaxy schema has been identified from the article where in order to facilitate the rapid filtering, drill-downs, and data queries optimized for the data warehouse, the system separates metrics into Fact Tables and descriptive criteria into Dimension Tables.

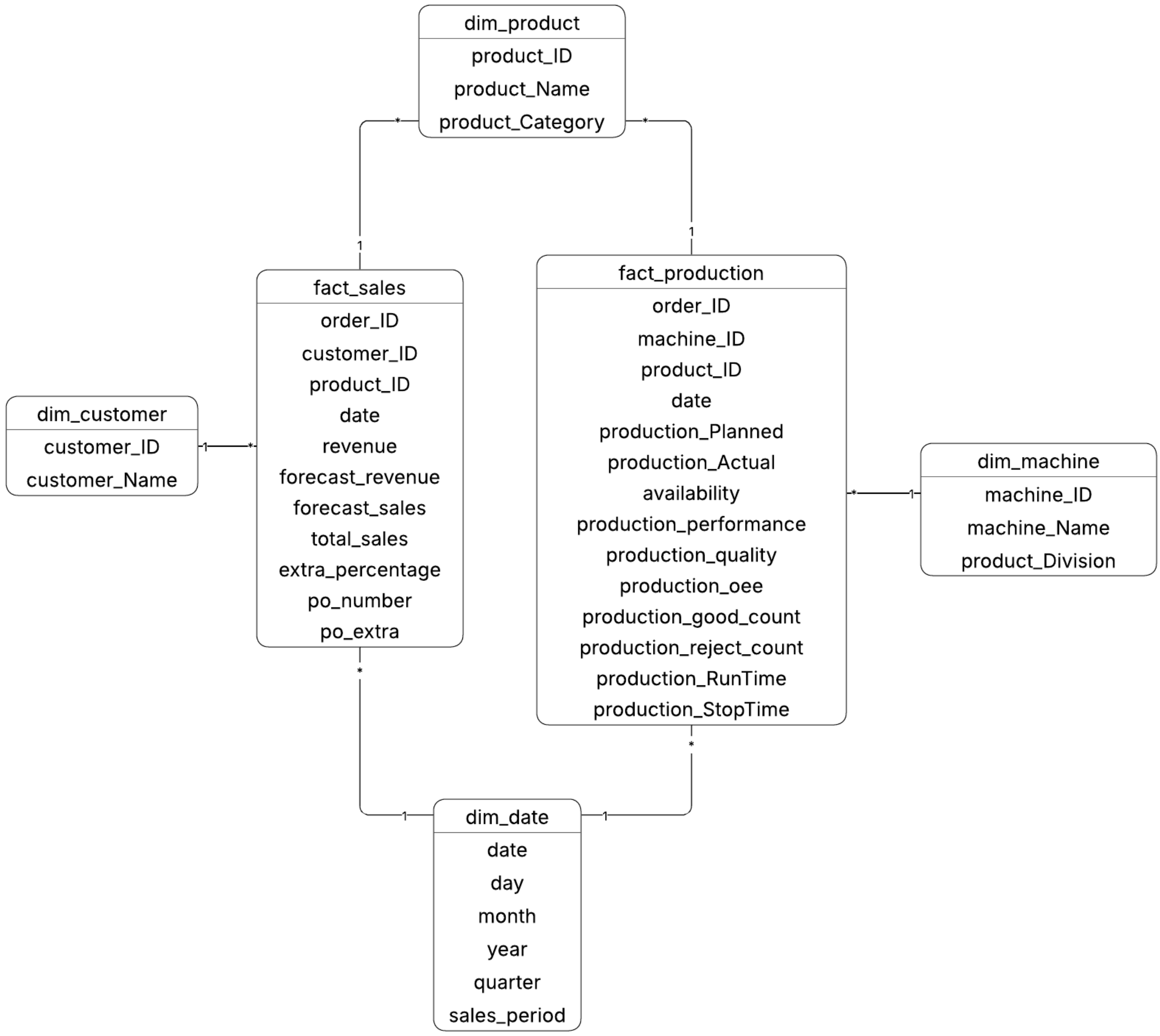

Figure 4: Galaxy Schema for Sales and Production for Google Data Studio